

Machine Learning — Day 02 Notes

How Does Machine Learning Learn?

Goal: Understand how ML learns patterns from examples and uses what it learned to make predictions.

1. Does a Machine Actually Think?

Not like a human. A Machine Learning system finds patterns in data and represents those patterns mathematically.

Example: If students who study more hours often pass, an ML system can learn a relationship between study hours and results from many examples.

2. The Basic Learning Process

Dataset → **Algorithm** → **Training** → **Model**

Dataset: Examples given to the system.

Algorithm: Method used to find patterns.

Training: Process of learning from data.

Model: Learned mathematical system used for prediction.

3. Learning from Examples

Study Hours	Result
1	Fail
2	Fail
4	Pass
5	Pass
6	Pass

The algorithm looks at examples and tries to find a useful pattern between the feature and the label.

For a new student with 7 study hours, the trained model can use the learned pattern to make a prediction.

4. What Does the Model Learn?

A model generally learns **parameters**. Parameters are values adjusted during training so the model can make better predictions.

A simple mathematical example is $y = wx + b$. Here x is the input, y is the prediction, and w and b are learned parameters.

You do not need the mathematics yet. The important idea is that training changes parameters so predictions improve.

5. What Happens When the Model Is Wrong?

During training, the model compares its prediction with the known label. The difference is measured using **error or loss**.

Prediction → Compare with actual answer → Calculate error → Adjust parameters → Try again

This process is repeated during training so the model can improve.

6. The Learning Loop

DATASET → ALGORITHM → PREDICT → COMPARE WITH LABEL → CALCULATE ERROR → ADJUST PARAMETERS → REPEAT

7. Training vs Prediction

Training	Prediction
The model learns.	The model is used.
Features + known labels	New features
Parameters are adjusted.	A prediction is produced.

8. Important Definitions

- **Algorithm:** Method used to learn patterns.
- **Training:** Process of learning from data.
- **Parameter:** A value learned and adjusted during training.
- **Loss/Error:** A measurement of prediction error.
- **Model:** The learned mathematical system used for prediction.
- **Prediction:** The output produced by a trained model for new input.

9. Day 2 Mental Map

Dataset → Algorithm → Training → Model → New Data → Prediction

Learning loop: **Prediction → Error → Adjust → Repeat.**

10. Problem Statement

Predicting Customer Churn: A streaming app wants to predict whether a customer will continue using the service or stop using it. It has past customer data such as watch time, number of logins, subscription duration, and whether the customer actually left.

Task: Explain how Machine Learning can learn from this past data and make a prediction for a new customer.

Day 2 — Key Idea

Machine Learning learns by studying examples, finding patterns, checking its mistakes, and adjusting its parameters to improve.